Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/20/2005
PI Name:	Rittmann, Bruce Ph.D.		
Project Title:	A Novel Membrane Microbial Fuel Cell for Energy and Resource Recovery from Solid Organic Wastes		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Arizona State University		
PI Address 1:	Center for Environmental Biotechnology		
PI Address 2:	Biodesign Institute		
PI Web Page:			
City:	Tempe	**State:**	AZ
Zip Code:	85287-5701	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/31/2004	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 08/31/2006 per PI (12/06) Project moved from Northwestern U in January 2005, per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lueptow, Richard M. (Northwestern University)		
Grant/Contract No.:	NNJ04HG05G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Membrane Microbial Fuel Cell (MMFC) is an innovative approach that could produce modest amounts of electricity while removing contaminants from the water and breaking down organic waste into water and other harmless compounds. The MMFC marries membrane technology with biotechnology. Bacteria catalytically oxidize the organic material in the MMFC to produce electrons that flow in an external circuit. Unlike a standard fuel cell, which operates at high temperature, the microbially catalyzed reactions in an MMFC occur at room temperature. Thus, the MMFC processes organic waste and wastewater in a way that helps produce pure water, generates electricity, and avoids extreme conditions in the spacecraft. To determine if and how this technology can be brought into practice, we have established the following objectives for the proposed study: a. Develop a bench-scale MMFC for laboratory testing b. Testing the MMFC with simple substrates to begin assessing its performance c. Establish the ecology and electrochemistry principles underlying the performance of an MMFC Although the MMFC is a very new approach, the potential advantages are enormous: processing organic waste to produce pure water and electricity with small size and no extreme conditions. Thus, the MMFC addresses several aspects of wastewater and solid waste in the AHST Critical Path Roadmap. To determine if and how this technology can be brought into practice, we have established the following objectives for the proposed study: a. Develop a bench-scale MMFC for laboratory testing b. Testing the MMFC with simple substrates to begin assessing its performance c. Establish the ecology and electrochemistry principles underlying the performance of an MMFC The Membrane Microbial Fuel Cell (MMFC) is an innovative approach that could produce modest amounts of electricity while removing contaminants from the water. The MMFC marries membrane technology with biotechnology. Bacteria catalytically oxidize the organic material and ammonium in the MMFC to produce electrons that flow in an external circuit. Unlike a standard fuel cell, which operates at high temperature, the microbially catalyzed reactions in an MMFC occur at room temperature. Thus, the MMFC processes organics and ammonium in wastewater in a way that helps produce pure water, generates electricity, and avoids extreme conditions in the spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the MMFC is still at the proof-of-concept state, its potential impacts could transform the way that society view materials now considered wastes. The MMFC will process biomass to produce pure water and electricity with high efficiency and without any air-pollution.
Task Progress:	1. We successfully oxidized acetate to generate electricity in a bench-scale MFC. 2. We created a simple MMFC by using a hollow-fiber membrane cathode (HFMC) in which the MFC cathode is attached directly onto a hollow-fiber membrane that supplied O2 for the cathode reaction. 3. We built a single-chamber MFC for conducting fundamental studies on the interaction of the biofilm biochemistry with the fuel-cell electrochemistry. 4. Using the single-chamber MFC, we demonstrated that the anode potential exhibits Monod-like kinetics, a feature we are exploring with mechanistic modeling. 5. We developed mechanistic mathematical models for how biofilm biochemistry interacts with fuel-cell electrochemistry.
Bibliography Type:	Description: (Last Updated: 06/15/2020)
Papers from Meeting Proceedings	Marcus A, Lueptow RM, Rittmann BE. "Preliminary Investigation of Generating Electricity from Wastewater via a Single-Compartment Microbial Fuel Cell." International Conference on Environmental Systems, Rome, Italy, SAE International, July 11-14, 2005. International Conference on Environmental Systems, Rome, Italy, July 11-14, 2005. SAE technical paper 2005-01-3112, https://doi.org/10.4271/2005-01-3112 , Jul-2005